

WASP-2b

R-0046

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_250821 · created 2026-07-09T05:04:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460908.7431 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.045
Transit depth (Rp/Rs)^2	2.34 +/- 1.38 [%]
Orbital Inclination (inc)	86.93 +/- 3.0
Airmass coefficient 1 (a1)	0.031 +/- 0.025
Airmass coefficient 2 (a2)	2.96 +/- 1.68
Scatter in the residuals of the lightcurve fit is	66.43 %
Best Comparison Star	#3 - [32, 205]
Optimal Aperture	2.05
Optimal Annulus	6.5
Transit Duration (day)	0.065 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.045 vs 0.1326 (deviation 0.45σ)
Duration fit vs published	0.065 d vs 0.0737 d (deviation 0.46σ)
Mid-time O–C	-19.7 min
Depth SNR	3.4
Residual scatter	66.43 %

Light curve

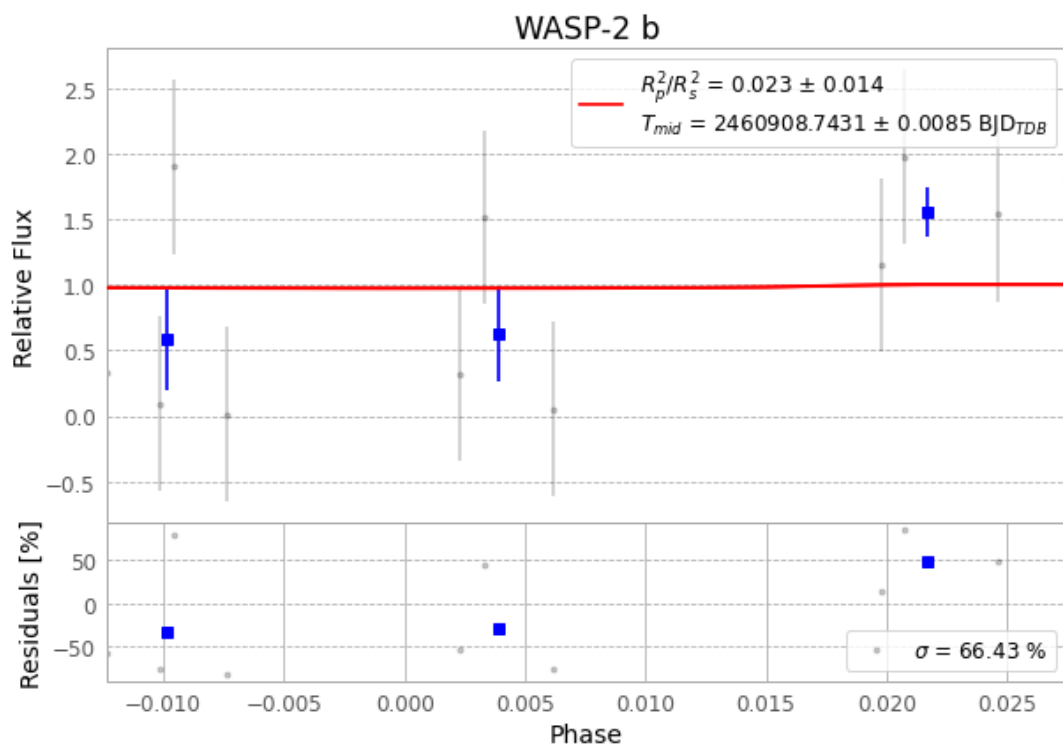


Figure 1 — FinalLightCurve_WASP-2 b_2025-08-20.png (SHA-256 7e363387f9ba6620...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 308e0aae1cb5275e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

66 FITS frames (43 MB), WASP-2250821045737.FITS → WASP-2250821075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250821.log (cbd2030d6ae6...), FinalLightCurve_WASP-2 b_2025-08-20.png (7e363387f9ba...), FinalLightCurve_WASP-2 b_2025-08-20.csv (841a1d4b035e...)

Compute resources

Wall time 139.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.